

Disaster Relief Resource Allocator

1. Introduction

The **Disaster Relief Resource Allocator** is a web-based system designed to efficiently manage and distribute relief resources during disasters. It helps authorities allocate food, medical aid, shelters, and rescue teams based on real-time needs and priority.

2. Objectives

- Efficient allocation of relief resources
- Real-time tracking of disaster-affected areas
- Priority-based distribution of aid
- Reduce response time during emergencies
- Ensure transparency and accountability

3. Technology Stack

Frontend: HTML , CSS , JavaScript , React

Backend: Node.js , Express.js

Database: PostgreSQL

Mapping & Visualization: Google Maps API / Leaflet.js

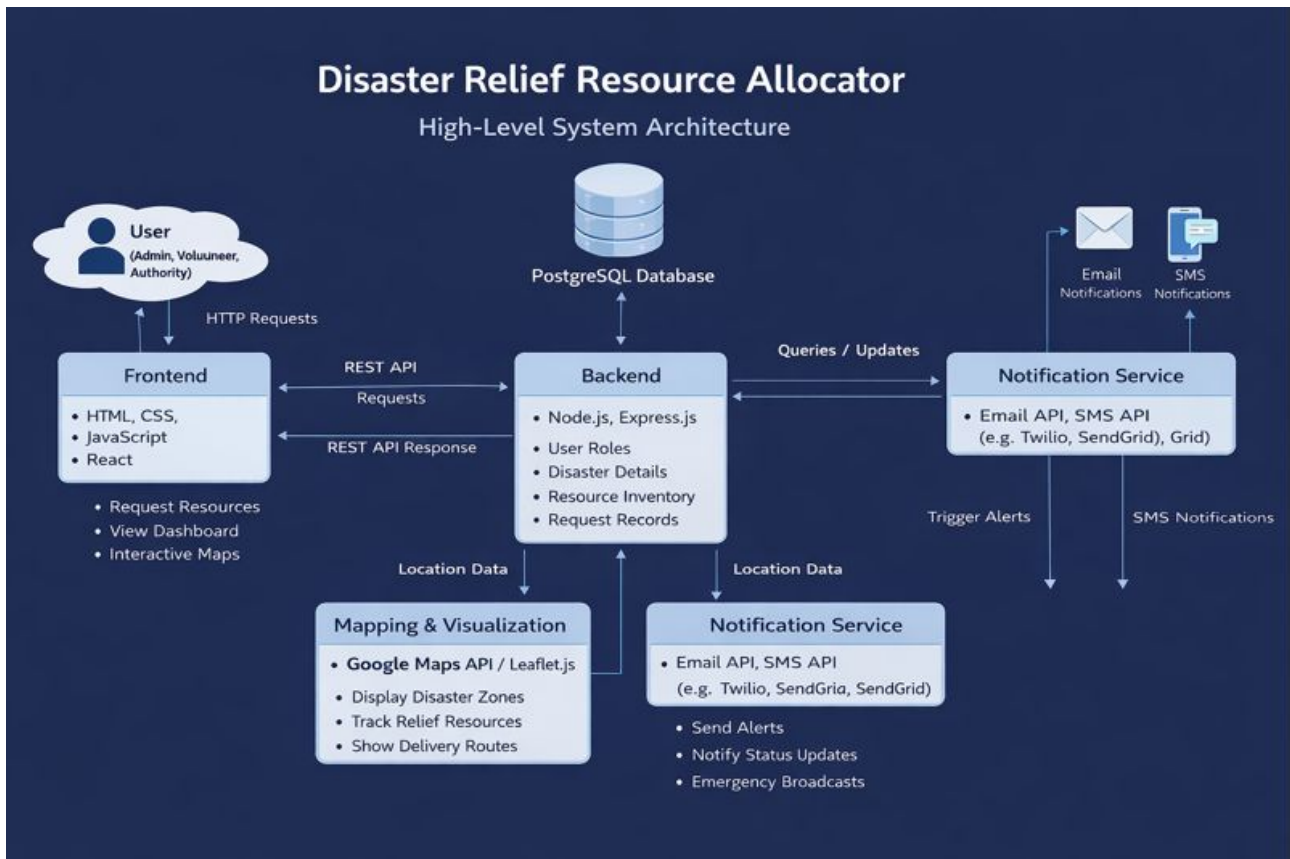
Notifications: Email / SMS API

4. High Level System Design

The system follows a **client-server architecture**. Relief administrators and field officers interact through a web interface. Requests are sent to the backend server, where allocation logic is processed. The database stores disaster data, resources, and requests. Maps and dashboards visualize affected regions and resource distribution.

High Level Architecture Components

- User Interface (Web Dashboard)
- Backend Application Server
- Database Server
- External APIs (Maps, Notifications)



5. Low Level System Design

5.1 User Management Module

This module handles registration and login of administrators, volunteers, and field officers. Role-based access control ensures users access only authorized features.

5.2 Disaster Management Module

This module records disaster details such as type, location, severity, and affected population. It allows real-time updates from field officers.

5.3 Resource Management Module

This module manages available resources like food, water, medicine, shelters, and rescue teams. It tracks quantity, location, and availability status.

5.4 Request & Allocation Module

Affected areas generate resource requests. The system allocates resources based on priority, availability, and proximity using predefined rules.

5.5 Tracking & Monitoring Module

This module tracks allocated resources and delivery status. It ensures transparency and helps administrators monitor relief operations.

5.6 Notification Module

Automated notifications are sent to volunteers, authorities, and relief centers regarding allocations, shortages, and urgent needs.

5.7 Database Design

The database includes collections such as:

- Users
- Disasters
- Resources
- Requests
- Allocations

MongoDB enables scalable and fast data handling during peak disaster situations.

6. Data Flow Diagram

The Data Flow Diagram represents how data flows from users to the system, how disaster information and resource requests are processed, and how allocation decisions are stored and communicated.

7. Security Measures

- User authentication and authorization
- Role-based access control
- Secure data storage
- Validation of disaster and resource inputs

8. Future Scope

- AI-based resource demand prediction
- Mobile application for field volunteers
- Integration with government disaster databases
- Drone and IoT-based real-time data collection

9. Conclusion

The **Disaster Relief Resource Allocator** provides a reliable and scalable solution for managing disaster relief operations. Its modular design ensures quick response, efficient allocation, and future extensibility during emergency situations.